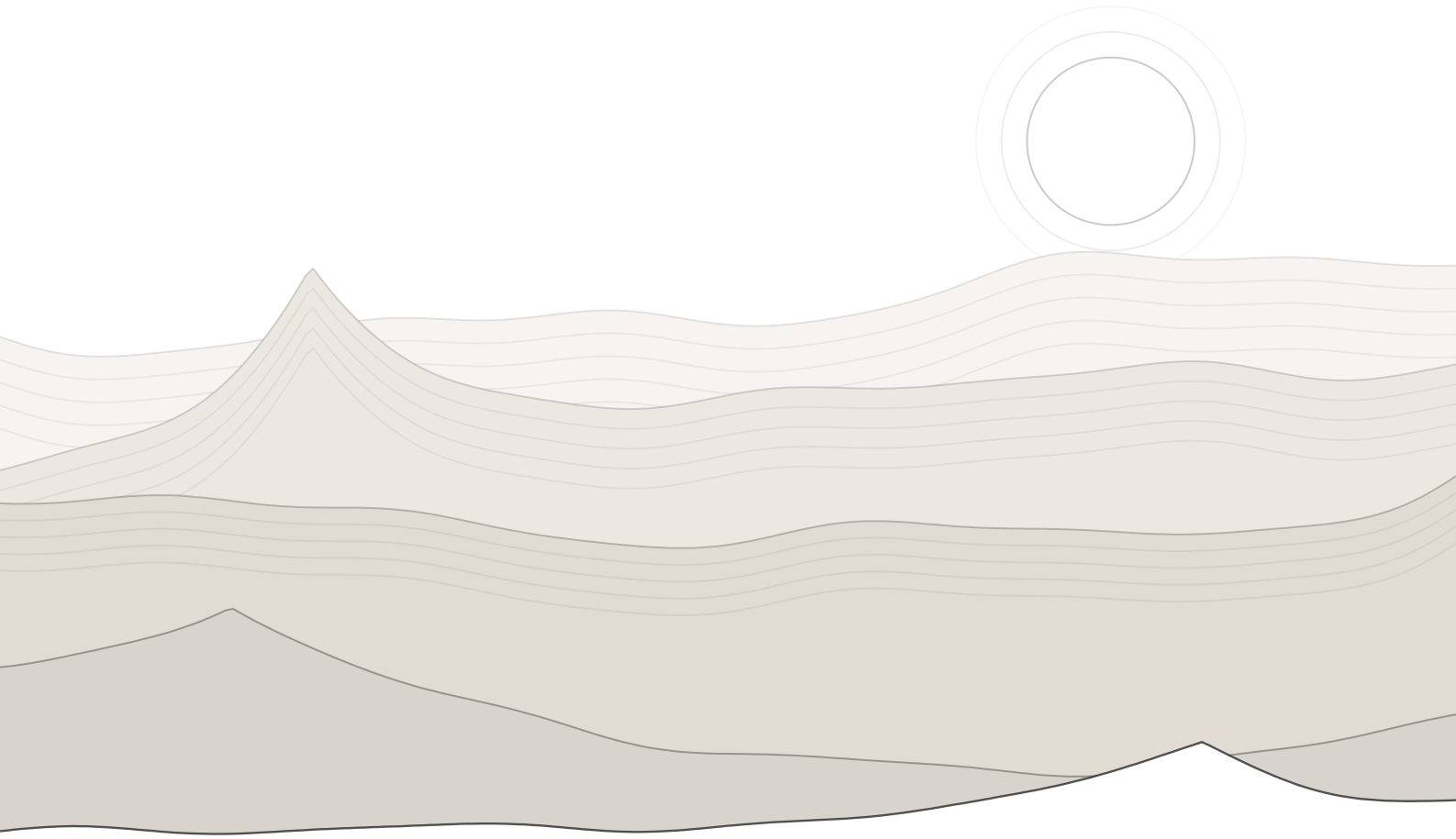
Matterhorn Capital Management

AMC 2026 · ROUND 1 · GROUP 3

Strategic Asset Allocation

Part A, the analysis of the allocation: outlook, universe, inputs, method, allocation and backtest (pages 3–14), code and portfolio (page 17). Part B, the checkpoint assignments: investment policy (page 2), robust approaches and backtest protocol (pages 15–16).



STARTING NAV	VOLATILITY TARGET	PUBLISHED
USD 100M	8% p.a.	18.10.26

Investment policy

CP1.1 · UNIVERSE

Long-only ETFs that track broad asset classes, listed in the US and traded in USD. No single stocks, no short positions, no leverage. We divide ETFs in three baskets by asset class, plus a T-bill ETF as the cash leg.

EQUITY · GROWTH	BONDS · DISINFLATION	REAL ASSETS · INFLATION	CASH LEG
VTI · VEA · IEMG	BND · BNDX · TIP	GLD · BCI	BIL

CP1.2 · OBJECTIVE AND RISK BUDGET

Objective: the highest Sharpe Ratio over the 1-month T-bill, holding up in all four macro regimes.
Strategy: equity is the growth engine, so we give it the largest share of risk; bonds and real assets are the brakes, performing when inflation falls and rises.

VOL. TARGET	EX-ANTE VOL.	1-MONTH 5% VAR	MAX ACCEPTABLE DRAWDOWN
8%	**8.13%**	**3.86%**	**−15%**
Range 7–9% p.a.	Last 60 months	= USD 3.86 million	Backtest −13.0% (2022)

CP1.3 · CONSTRAINTS, WITH REASONS

CONSTRAINT WE IMPOSE	REASON (TODAY'S MODEL ERRORS)
Concentration: we allocate risk, not capital. Rules: • Equity basket ≤ 70% of the risk, bonds + commodities ≥ 30% (split equally) • Equal risk contribution within each basket	**Estimation error.** Over five years, the mean return of a 15%-vol ETF has a standard error of about 7 pp, so we use no method based on expected returns (Maximum Sharpe puts 35% in gold). **ETF-count bias.** Equal weight, 1/σ and ERC on all ETFs let the number of ETFs per asset class set its risk. **Equity risk** kept lower than in a traditional 60/40 portfolio (~90%).
Regions: • Each equity and bond region must be allocated capital • ETFs are weighted by risk, not by pure market-cap weight • No position below 2%	**Model error: concentration.** We limit the risk of holding concentrated US AI mega caps, with analysts questioning circular financing among them [17]. The top ten stocks make up 36.1% of VTI [5] and roughly 25% of the world equity market [19].
Liquidity: • ETF AUM ≥ USD 1 billion and reasonably high trading volume	**Model error:** The optimiser assumes trading at the close is feasible and has no cost.
Cash leg: • Allocate BIL only if portfolio volatility > 9% (upper bound of the target volatility range)	**Keeping cash.** Cash earns −4% excess return, while BIL has near-zero volatility and earns about 0% excess return.

CP1.4 · RISK CONTRIBUTIONS

ETF	WEIGHT	RISK CONTR.	ETF	WEIGHT	RISK CONTR.
VTI	10.90%	15.21%	TIP	15.26%	8.84%
VEA	10.24%	17.42%	GLD	11.15%	15.49%
IEMG	10.23%	17.36%	BCI	12.40%	9.51%
BND	13.33%	8.34%	BIL	0.01%	0.00%
BNDX	16.48%	7.83%	**Total**	**100.00%**	**100.00%**

We chose a 50/25/25 risk allocation to our baskets. **No position is above half of the portfolio risk** (largest: VEA, 17.4%). Equity holds 31.4% of capital and 50.0% of risk; bonds 45.1% and 25.0%; real assets 23.5% and 25.0%.

Market outlook

Our reading of the macro picture is that 2026 has been a year of stagflation, and that economists expect a return to goldilocks in 2027. Markets agree that inflation will fall, but not on how: economists expect the Fed to cut as inflation falls, while fed funds futures price about 3.4 further hikes, as if inflation will fall only because the Fed keeps rates high. The two paths favour different asset classes, and we cannot tell which one we are on. Hence, we do not build the portfolio around a single scenario, but hold assets that perform in each of them.

1.1 · GROWTH AND INFLATION

We classify each economy into the four regimes defined by the direction of growth and inflation, using the Bloomberg consensus (ECFC). Between Q1 and Q3 2026 inflation rose in every region. In the United States and the emerging economies growth slowed at the same time, which places both in the stagflation quadrant (growth slowing, not contracting): US CPI rose from 2.7% to 3.4% and real GDP growth fell from 2.6% to 1.8%. In the other developed economies growth rose as well, so they are in the overheating quadrant. For 2027, the consensus expects inflation to fall back to 2–3% in every region, with growth broadly steady: a return to goldilocks in the United States and the emerging economies. [1]

TABLE 1 · CONSENSUS FORECASTS AND MACRO REGIMES

REGION	GDP Q1 26	GDP Q3 26	GDP Q4 27	CPI Q1 26	CPI Q3 26	CPI Q4 27
United States	2.6%	1.8%	2.1%	2.7%	3.4%	2.1%
Developed economies	1.8%	2.1%	1.9%	3.1%	3.7%	2.5%
Emerging economies	4.8%	4.0%	4.4%	3.0%	3.4%	2.9%

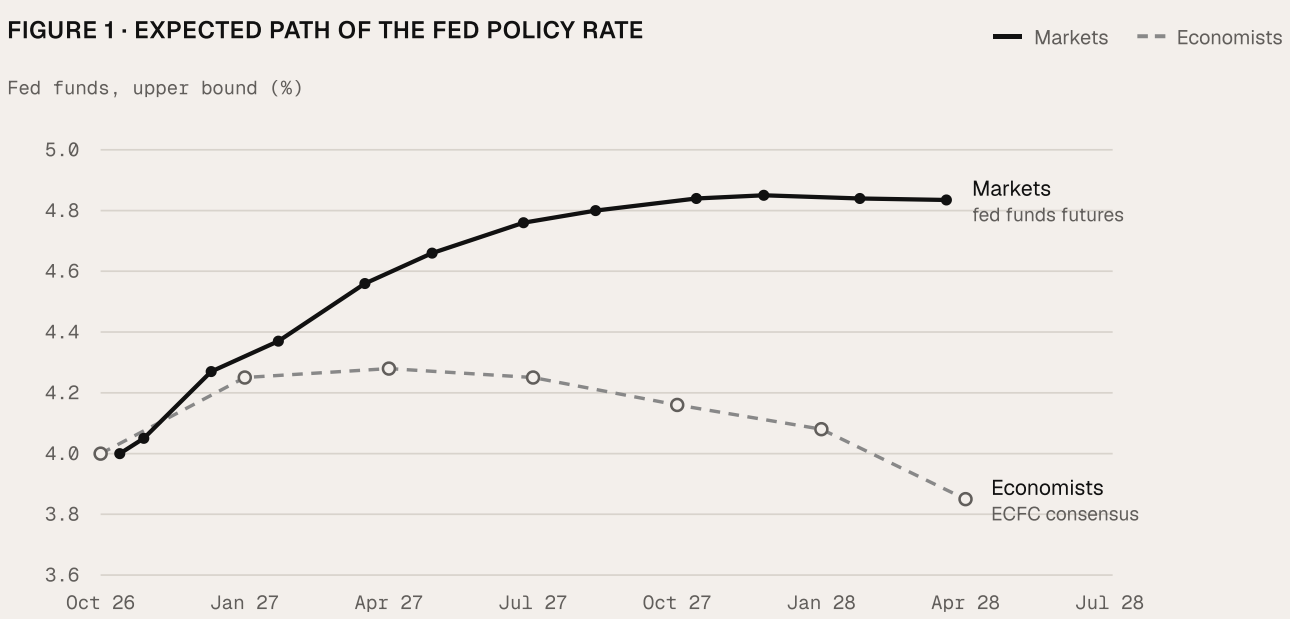
Real GDP and CPI, year-on-year. Q3 2026: consensus estimate; Q4 2027: consensus forecast. [1]

Behind these numbers, the US economy is slowing gently rather than contracting. Unemployment is stable at about 4.2% and payroll growth has eased to about 60,000 jobs a month, while private investment remains the strongest component of demand, having grown at an annualised 7.4% in the third quarter. Inflation is still the main concern, with core PCE at 3.3%. Europe continues to grow slowly, at 1.3–1.6%, whereas the emerging economies combine rising growth with monetary easing, as their central banks are expected to cut rates from 6.3% to about 5.0% by early 2028. [1]

Monetary policy: economists versus markets

After the September hike the Fed funds target range stands at 3.75–4.00%, and the path of the policy rate is where economists and markets disagree most. Economists expect one further hike followed by cuts that would bring the rate to 3.85% by early 2028. Markets expect the opposite: fed funds futures, whose prices reflect the rate investors expect, imply a rate of about 4.84% by the end of 2027, roughly three to four further quarter-point hikes. [1, 2]

FIGURE 1 · EXPECTED PATH OF THE FED POLICY RATE



Markets: implied upper bound from fed funds futures. Economists: Bloomberg ECFC consensus. [1, 2]

IF ECONOMISTS ARE RIGHT

Bonds rally and growth assets benefit from lower rates.

IF MARKETS ARE RIGHT

Inflation falls, but real rates stay high: bonds, TIPS included, and equities come under pressure.

We read this gap as the clearest evidence that the path of rates, and with it which asset class performs best, is genuinely uncertain, and it is the starting point of our allocation.

1.3 · RATES, INFLATION PRICING, CREDIT AND THE DOLLAR

INDICATOR	1 YEAR AGO	LATEST
US Treasury 1-month yield	4.20%	4.13%
US Treasury 2-year yield	3.60%	4.80%
US Treasury 10-year yield	4.14%	5.24%
US Treasury 30-year yield	4.72%	5.60%
5-year breakeven inflation	2.39%*	2.34%
10-year real yield (TIPS)	1.78%	2.87%
US investment-grade spread (OAS)	0.75%	0.82%
US high-yield spread (OAS)	2.84%	3.15%
VIX	16.3	15.4
EUR/USD, consensus Q3 2026 → Q4 2027	1.13	1.17

*Table 2 · Key market indicators. Treasury yields and 1-year-ago real yield (8 Oct 2025): U.S. Treasury par and real yield curves [3]. Latest breakeven, real yield, spreads and VIX: FRED (T5YIE, DFII10, BAMLC0A0CM, BAMLH0A0HYM2, VIXCLS) [4]. *1-year-ago breakeven (9 Oct 2025) calculated as 5Y nominal 3.74% − 5Y real 1.35%. 1-year-ago VIX: close of 8 Oct 2025 [12]. EUR/USD [1].*

The indicators in Table 2 point to four features of the current market that shape our universe. The Treasury curve has shifted up by 80–130 basis points from the one-year maturity onwards, so that bonds once again offer a meaningful income. The inflation priced by the market, 2.34% a year over five years, is in line with the 2.3% economists expect by early 2028 and below the 3%+ they expect for the coming quarters, so inflation protection is not expensive. Over the year the 10-year real yield rose by about 109 bp (1.78% → 2.87%) while five-year inflation expectations were almost unchanged (2.39% → 2.34%): the rise in nominal yields came from real rates, not from inflation expectations. Credit spreads remain tight, so that credit offers little compensation for its risk, and the consensus expects a mild weakening of the dollar, which would support non-US assets measured in USD.

How we built the universe

01

No bet on a single regime

Given the outlook above, we hold assets that perform in each of the four regimes rather than picking one.

02

Cheap, broad market exposures

The policy portfolio drives most of the risk and return, so we buy beta passively: broad market-cap indices at low cost. Sector and factor bets are left for the tactical rounds.

03

No duplicates

No two ETFs track the same market. One small overlap remains, gold (see section 03).

EQUITY	FIXED INCOME	REAL ASSETS	RISK TOOL
VTI · VEA · IEMG	BND · BNDX · TIP	GLD · BCI	BIL

The ETFs and why we chose them

Each ETF has a role in our strategy: equity is the growth engine; bonds are the brake when inflation falls, real assets when it rises. BIL is a tool to manage the risk budget, not a strategic position.

3.1 · EQUITY

Equity earns the equity risk premium, the best-documented source of long-run excess return. We hold three regional ETFs weighted by risk, not by market capitalisation. A market-cap allocation would be concentrated: 64% in the US (MSCI ACWI [6]), and much of it in a few AI-linked mega caps, as VTI's top ten holdings are 36.1% of the fund [5]. Weighting by risk limits the US share: the US is 35% of our equity.

VTI Vanguard Total Stock Market ETF

US equity · Morningstar US Total Market Index · listed May 2001 · 3,519 stocks · TER 0.03% [5]

WHY WE CHOSE IT	The largest and deepest equity market, with the strongest earnings growth of the last decade, led by technology and AI.
WHY NOT SPY / VOO	SPY and VOO hold the 500 largest companies; VTI adds mid and small caps, more sensitive to domestic growth and falling rates, and is slightly less concentrated in mega caps.
RISKS WE ACCEPT	High valuations and mega-cap concentration; with the 10-year yield at 5.24%, the equity risk premium is thin. Risk weighting and the two non-US ETFs limit both.

VEA Vanguard FTSE Developed Markets ETF

Developed ex-US equity · FTSE Developed All Cap ex US Index · listed July 2007

WHY WE CHOSE IT	Developed ex-US markets hold more financials, industrials and healthcare and less technology, so they react differently to the same shocks.
WHY NOT VXUS	It would merge in emerging markets, which we size separately.
RISKS WE ACCEPT	Slow growth in Europe (1.3–1.6% [1]) and exporters exposed to tariffs. The fund is unhedged: it gains if the dollar weakens, as the consensus expects, and loses if it strengthens.

IEMG iShares Core MSCI Emerging Markets ETF

Emerging markets equity · MSCI Emerging Markets Investable Market Index · listed October 2012

WHY WE CHOSE IT	It completes the coverage of world equity and gives access to the semiconductor supply chain (Taiwan, Korea).
RISKS WE ACCEPT	Higher volatility; geopolitical and regulatory risk in Taiwan and China, about 29% and 20% of the index [20]; losses when the dollar and US rates rise. South Korea is held twice: Developed for FTSE since 2009 [7], so in VEA, but still Emerging for MSCI in June 2026 [8].

3.2 · FIXED INCOME

Bonds are the brake when inflation falls: in a recession with falling inflation, central banks cut and bond prices rise. They also lower the volatility of the portfolio, so that it stays within the 7–9% budget. We hold three sources of return: US nominal rates, non-US rates and US real rates.

BND Vanguard Total Bond Market ETF

US investment-grade bonds · Bloomberg US Aggregate Float Adjusted Index · listed April 2007

WHY WE CHOSE IT Treasuries, agency MBS and IG corporate bonds at intermediate duration. With the 10-year yield at 5.24%, bonds pay a meaningful income, and the fall to about 4.6% that economists expect by end-2027 [1] would add a capital gain.

WHY NOT TLT With a deficit of about 6.5% of GDP and debt at 105% [1], Treasury supply may keep long yields high: intermediate duration keeps part of the protection with less of this risk. High yield pays little (spread 3.15%) for a risk that rises when equities fall.

RISKS WE ACCEPT In our sample BND's correlation with VTI is 0.50 (Table 4): when inflation is the problem, bonds and equities can fall together, as in 2022. Futures also price further Fed hikes.

BNDX Vanguard Total International Bond ETF

International IG bonds, USD-hedged · Bloomberg Global Aggregate ex-USD Float Adjusted RIC Capped Index · listed June 2013

WHY WE CHOSE IT It extends the bond basket beyond US rates and does not depend on US Treasury supply, the risk that keeps us out of long US bonds. Its duration, about 6 years, keeps the basket at intermediate duration.

WHY NOT IGOV IGOV is unhedged: currency moves would add equity-like volatility to our defensive assets.

RISKS WE ACCEPT Its correlation with BND is 0.89 (Table 4), so it diversifies little in a global rate shock, which would push yields up everywhere at once.

TIP iShares TIPS Bond ETF

US inflation-linked Treasuries · listed December 2003

WHY WE CHOSE IT Its risk is mostly real-rate duration, which is why it sits in the bond basket: its correlation with BND is 0.90 (Table 4). It also partly protects against inflation, and not at a high price: the 5-year breakeven (2.34%) is in line with the 2.3% CPI economists expect by early 2028 [1, 4], and the 2.87% real yield pays even if inflation returns to target.

WHY NOT STIP STIP holds only 0–5 year maturities; with real yields near 2.9%, we want some real-rate duration.

RISKS WE ACCEPT TIP loses when real yields rise, even with high inflation, as in 2022; and if inflation falls faster than priced, it lags nominal bonds such as BND.

3.3 · REAL ASSETS

Real assets are the brake when inflation rises, the case in which equities and nominal bonds can fall together. They are also the least correlated with US equity: 0.14 for GLD and 0.09 for BCI with VTI (Table 4).

GLD SPDR Gold Shares

Gold · physical gold bullion · listed November 2004

WHY WE CHOSE IT Gold has no credit or counterparty risk and has historically held its value in inflation and in crises. It reacts to monetary and geopolitical stress more than to the real economy.

RISKS WE ACCEPT No income, with real yields near 2.9%, and a strong rally behind it: the price hit a record USD 5,405/oz in January 2026, followed by a notable correction, and averaged USD 4,873/oz in Q1 2026 [9].

OVERLAP WITH BCI Gold is also 14.9% of the Bloomberg Commodity Index [18], so our gold is about 13.0% of the capital: 11.15% in GLD plus 1.8 pp through BCI. We accept this, as the overlap adds less than 2 pp.

BCI Bloomberg All Commodity Strategy K-1 Free ETF

Broad commodities · Bloomberg Commodity Index Total Return · listed March 2017

WHY WE CHOSE IT Commodities gain when inflation comes from supply shocks, the case in which stocks and bonds both lose, as in 2022. They react to the real economy, so they complement gold (correlation 0.31, Table 4). The total-return index also earns the T-bill rate on the collateral.

WHY NOT DBC BCI follows the Bloomberg Commodity Index: 23 commodities, no sector above 33% and no single commodity above 15% [11]. DBC is about half energy and more expensive [10].

RISKS WE ACCEPT Roll costs in markets in contango, and sharp falls in a recession, when demand for raw materials collapses.

BIL SPDR Bloomberg 1-3 Month T-Bill ETF

3.4 Risk-management tool · US Treasury bills, 1-3 months · listed May 2007

ROLE Not a strategic position. BIL is used only if the risk budgets give an ex-ante volatility above the 9% cap: the risky weights are scaled down to 9% and the rest goes to BIL, as in the 70/15/15 budget (Table 5). Our SAA, at 8.13%, is within the band, so BIL plays no role (0.01%, an optimiser residual).

WHY NOT CASH The Sharpe Ratio is measured on returns in excess of the 1-month T-bill (4.13%) [3]. Cash earns nothing, so its excess return is about −4% a year; BIL's is close to zero.

The inputs

All inputs are estimated on the same 60 complete months, **October 2021 – September 2026**, as the guidelines require. Returns are monthly total returns in USD, computed from Bloomberg adjusted prices, which include dividends [13]. Means and variances are annualised ×12, volatilities ×√12. BCI only started trading in 2017, so for the whole sample we use the index it tracks, the Bloomberg Commodity Index Total Return (BCOMTR), before the fund's fees. In this way, the estimation window and the backtest use the same series. The risk-free rate is the 1-month T-bill (FRED DGS1MO) [14], the rate on which the Sharpe Ratio of the challenge is computed.

TABLE 3 · VOLATILITY AND MEAN

ETF	VOLATILITY	HIST. EXCESS RETURN*
VTI	15.24%	8.97%
VEA	15.67%	7.02%
IEMG	16.28%	5.86%
BND	6.76%	−4.25%
BNDX	5.57%	−3.74%
TIP	6.00%	−3.87%
GLD	17.08%	14.58%
BCI†	15.36%	8.10%

**Mean monthly return minus the T-bill, ×12. Shown, not used by our method. †BCOMTR index.*

TABLE 4 · CORRELATIONS

	VTI	VEA	IEMG	BND	BNDX	TIP	GLD	BCI
VTI	1	0.82	0.72	0.50	0.53	0.53	0.14	0.09
VEA	0.82	1	0.81	0.64	0.58	0.62	0.42	0.20
IEMG	0.72	0.81	1	0.56	0.47	0.53	0.44	0.24
BND	0.50	0.64	0.56	1	0.89	0.90	0.46	−0.06
BNDX	0.53	0.58	0.47	0.89	1	0.85	0.41	−0.13
TIP	0.53	0.62	0.53	0.90	0.85	1	0.47	0.12
GLD	0.14	0.42	0.44	0.46	0.41	0.47	1	0.31
BCI	0.09	0.20	0.24	−0.06	−0.13	0.12	0.31	1

−1 +1: blue to red through white at 0, as in the notebook. Monthly returns, Oct 2021 – Sep 2026.

4.1 · WHAT THE CORRELATIONS TELL US

Each basket behaves almost as one asset. The three equity ETFs correlate 0.72–0.82 and the three bond ETFs 0.85–0.90, so there is little diversification inside a basket. Diversification comes from combining the baskets, which is why our method sets the risk of each basket (section 05).

Stocks and bonds have moved together. Their correlation is 0.47–0.64 over a window that includes 2022. Bonds still lower the volatility, but they hedge equities less than they did before 2021 (Class 3, Part II). For this reason we do not rely on bonds alone for protection.

Gold and commodities are the real diversifiers. Their correlations with the other ETFs are at most 0.47, and commodities are uncorrelated with bonds (−0.13 to 0.12). This is the statistical side of the role that we give to real assets in section 03.

4.2 · WHY WE DO NOT USE THE EXPECTED RETURNS

The historical excess returns describe the last five years, not the next ones. Bonds show −4% a year because yields rose sharply in 2022, the same rise that now gives bonds a yield above 5%. Gold shows +14.6% because of a record rally. With five years of data, the standard error of the mean of an ETF with a 15% volatility is 15% / √5 ≈ 6.7 percentage points, larger than most of the means themselves. An optimiser using these numbers puts a third of the capital in gold and nothing in bonds (section 05). Our weights therefore depend only on the volatilities and the correlations, which are estimated much more precisely.

The method: risk budgets by basket

Our weights come from a two-level risk-budgeting rule that uses only the covariance matrix: equal risk per ETF inside each basket (ERC) and, across the baskets, **50% equity, 25% bonds, 25% real assets**. Long-only, no leverage, ex-ante volatility between 7% and 9%.

5.1 · HOW THE WEIGHTS ARE COMPUTED

Step 1 · inside the basket

ERC weights, so that $w_i(Vw)_i$ / $w'Vw$ is the same for every ETF of the basket. Unlike 1/σ, ERC uses the correlations.

Step 2 · across baskets

Basket weights that bring each basket's share of risk closest to 50 / 25 / 25, with $w \geq 0$, $\Sigma w \leq 1$ and $7\% \leq \sigma \leq 9\%$.

Step 3 · combine

ETF weight = basket weight × weight inside the basket; BIL takes the rest only when the 9% cap binds. Solved numerically (SLSQP, scipy).

5.2 · WHY A RISK-BASED METHOD

The asset class, not the ETF count. Equal weight, risk parity 1/σ and ERC on single ETFs allocate by ETF, so the result depends on how many ETFs each asset class has: under ERC, three bond ETFs out of eight take 3/8 of the risk. Our method budgets the risk by asset class.

Estimation error. Mean–variance and maximum Sharpe maximise the in-sample Sharpe Ratio, but with means this noisy (section 04) the weights overfit: maximum Sharpe puts 35% in gold.

Regimes. Equities pay in goldilocks, bonds in a recession with falling inflation, real assets under inflation (Class 3). Fixing the risk of each basket means no regime dominates the portfolio, and the next regime is uncertain (section 01).

5.3 · WHY 50 / 25 / 25

Equity is the growth engine: its premium, the best-documented long-run premium, rests on growth and earnings [16]; bonds and commodities are the brakes, in falling and rising inflation. We therefore give equity the largest share of the risk, at most 70%, and the brakes at least 30%, split equally.

Equal risk across the three baskets gives an ex-ante volatility of 7.39%, near the bottom of our band. We tested three budgets with more equity risk (Table 5) and chose 50/25/25, the closest to the 8% target on the in-sample estimate. Only then did we look at the hold-out: its Sharpe Ratios are practically equal (0.65–0.66, within the noise), while volatility and drawdown keep the in-sample ranking. The budget controls how much risk we take, not the return per unit of risk. A 60/40 portfolio has about 90% of its risk in equity, so even our 70% cap is more balanced.

TABLE 5 · THE FOUR EQUITY BUDGETS

RISK BUDGET EQUITY / BONDS / REAL	EX-ANTE VOL.	VOL., HOLD-OUT	MAX DD, HOLD-OUT	SHARPE, HOLD-OUT
33 / 33 / 33	7.39%	5.86%	−10.4%	0.65
50 / 25 / 25 · our SAA	**8.13%**	**6.94%**	**−11.2%**	**0.66**
60 / 20 / 20	8.71%	7.79%	−12.8%	0.66
70 / 15 / 15	9.00%	8.41%	−14.2%	0.66

Hold-out: Jul 2013 – Sep 2021, months not used for the weights. At 70/15/15 the 9% cap binds and the rest goes to BIL.

The methods we compared

We computed eleven allocations on the same inputs and rules (long-only, no leverage, rest in BIL, ex-ante volatility in 7–9%). Our candidates were the four basket budgets, on economic grounds (section 05). Equal weight and risk parity 1/σ are the benchmarks of Checkpoint 2. The other methods we computed for comparison: they allocate risk ETF by ETF or rely on expected returns, and minimum variance stays at the bottom of our volatility band.

TABLE 6 · WEIGHTS OF THE ELEVEN ALLOCATIONS

METHOD	VTI	VEA	IEMG	BND	BNDX	TIP	GLD	BCI	BIL	VOL.
CANDIDATES — BASKET METHOD										
Basket 33/33/33	7.1	6.7	6.7	15.7	19.4	18.0	12.4	13.8	0.1	7.39
Basket 50/25/25 · SAA	**10.9**	**10.2**	**10.2**	**13.3**	**16.5**	**15.3**	**11.1**	**12.4**	**0.0**	**8.13**
Basket 60/20/20	13.5	12.7	12.7	11.7	14.5	13.4	10.2	11.4	0.0	8.71
Basket 70/15/15	15.8	14.8	14.8	9.4	11.6	10.7	8.6	9.5	4.7	9.00
BENCHMARKS OF CHECKPOINT 2										
Equal weight	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	0.0	8.78
Risk parity 1/σ	8.1	7.9	7.6	18.3	22.2	20.6	7.2	8.1	0.0	7.23
OTHER METHODS — FOR COMPARISON										
Minimum variance	9.8	9.2	9.2	11.3	11.6	11.4	9.8	11.0	16.7	7.00
Maximum Sharpe	26.0	0.0	0.0	0.0	0.0	0.0	35.3	14.6	24.0	8.61
Mean–variance at 8%	24.0	0.0	0.0	0.0	0.0	0.0	32.9	13.5	29.5	8.00
ERC, 8 ETFs	8.2	6.7	6.8	16.7	21.8	17.9	8.0	13.7	0.2	7.13
Max. diversification	16.4	0.0	0.0	5.3	41.4	0.0	10.6	26.4	0.0	7.00

Weights and ex-ante volatility in %, Oct 2021 – Sep 2026. Equal weight and 1/σ are fully invested by definition; the others are solved under the 7–9% band.

WHY WE DID NOT CHOOSE THE OTHERS

OTHER CANDIDATES
33/33/33, 60/20/20, 70/15/15

33/33/33 was our starting point: equal risk per basket gives 7.39%, near the bottom of the band, so we moved risk towards equity. 60/20/20 and 70/15/15 overshoot the 8% target (8.71% and 9.00%, the last only with 4.71% in BIL). All four have practically the same hold-out Sharpe Ratio (Table 5).

BENCHMARKS
Equal weight, risk parity 1/σ

They allocate by ETF and let the ETF count decide; equal weight also ignores the risk of each ETF, and 1/σ the correlations. In the hold-out, 1/σ had a higher Sharpe Ratio than ours (0.78 against 0.66), probably because stocks and bonds often moved in opposite directions in 2013–2021, which favours its 61% in bonds. We did not adopt it: the hold-out evaluated our choice, it did not make it.

OTHER METHODS
Maximum Sharpe, free and at 8%

The only rules that use expected returns, and it shows: a third of the capital in gold, a quarter in US equity, no bonds, no non-US equity and 24–30% in BIL. They load on the winners of the last five years (Class 3).

OTHER METHODS
Minimum variance

Too little risk for our objective: it stops at the 7% floor of the band and leaves 16.7% in BIL. It is also not unique: every portfolio at exactly 7% is optimal, so the weights depend on where the solver starts.

OTHER METHODS
Maximum diversification

It assumes equal Sharpe Ratios across ETFs and loads on the least correlated ones: 41% in BNDX and 26% in commodities, no non-US equity. Those correlations are specific to our sample: stocks and bonds moved together in 2021–2026, unlike in 2013–2021, so the weights would not be robust.

OTHER METHODS
ERC on the 8 ETFs

The ETF count becomes a hidden risk budget: three bond ETFs out of eight receive 3/8 of the risk. It also stays near the bottom of the band (7.1%).

The allocation

EX-ANTE VOLATILITY	1-MONTH 5% VAR	POSITIONS	LARGEST RISK CONTR.
8.13%	3.86%	8 ETFs	17.4%
Band 7–9%	USD 3,860,465	Three baskets · BIL not held	VEA

TABLE 7 · CAPITAL WEIGHT AND RISK CONTRIBUTION

Capital Risk

ETF	CAPITAL	RISK	
EQUITY · 31.37% OF CAPITAL · 50.0% OF RISK			
VTI	10.90%	15.21%	
VEA	10.24%	17.42%	
IEMG	10.23%	17.36%	
BONDS · 45.07% OF CAPITAL · 25.0% OF RISK			
BND	13.33%	8.34%	
BNDX	16.48%	7.83%	
TIP	15.26%	8.84%	
REAL ASSETS · 23.54% OF CAPITAL · 25.0% OF RISK			
GLD	11.15%	15.49%	
BCI	12.40%	9.51%	
BIL	0.01%	0.00%	Cash leg · residual, goes to equity (section 09)

6.1 · CAPITAL VERSUS RISK

Bonds take 45% of the capital but only 25% of the risk; equity takes 31% of the capital and 50% of the risk. A dollar in equity moves the portfolio about 2.6 times as much as a dollar in bonds (volatility 15–16% against 6–7%). The capital weights therefore look defensive, but the risk is split evenly between growth and protection, by design.

6.2 · WHY CONTRIBUTIONS TO PORTFOLIO RISK ARE NOT EQUAL INSIDE THE BASKETS

Inside each basket, contributions differ because step 1 equalises risk within the basket alone, ignoring each ETF's covariance with the other baskets; the basket totals match the budgets exactly.

6.3 · VOLATILITY AND VALUE AT RISK

The ex-ante volatility, $\sqrt{(w'Vw)} = 8.13\%$, is inside the band and close to the target. Under normality, the 1-month 5% VaR is $1.645 \times 8.13\% / \sqrt{12} = 3.86\%$: in one month out of twenty we expect to lose more than USD 3.86 million. We read this as a reference, not a ceiling. In the backtest the worst month of these weights lay far beyond the normal VaR, so the true left tail is heavier. Realised volatility can also differ from 8.1%: the estimation window includes 2022, while in 2013–2021, when stocks and bonds often moved in opposite directions, the same weights had a volatility of 6.9%.

The backtest

We hold the SAA weights fixed and rebalance them to target at the end of every month. The sample is July 2013 – September 2026 (159 months): it starts when every ETF exists, the limit being BNDX (June 2013), whose index is not available to us before then. We ignore transaction costs, and count them among the reasons for the discount (CP3.4). Because the weights are estimated on the last 60 months, a single backtest would partly test the weights on the data that produced them. We therefore run three tests, each with its own question and limit:

Full sample
Our weights on all 159 months. Reference only: the last 60 months are the estimation window, so they are in-sample.

Hold-out · the test we rely on
Our weights on the 99 months before the estimation window, never seen by the optimiser. No in-sample fitting, but look-ahead (2021–2026 weights on 2013–2021). It tests the mix we hold, not the method or re-optimising.

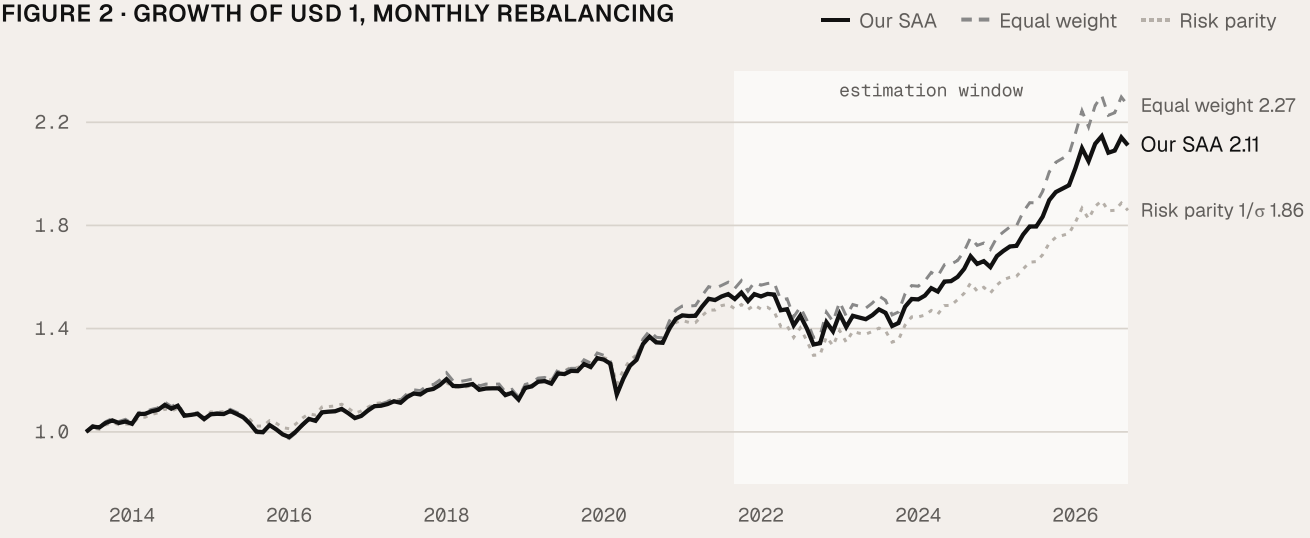
Split
The rule estimated once on the first 60 months and held fixed on the following 99, as our SAA is held fixed. No look-ahead, but it tests the method: its weights are not ours.

TABLE 8 · PERFORMANCE AND RISK OF THE SAA

	FULL SAMPLE	HOLD-OUT	SPLIT
Period	07.13-09.26	07.13-09.21	07.18-09.26
Return, annualised	5.92%	5.28%	9.75%
Cumulative return	111.1%	51.5%	113.2%
Volatility, annualised	7.39%	6.94%	10.35%
Sharpe Ratio (t-stat.)	0.55 (2.00)	0.66 (1.91)	0.68 (1.94)
Sortino Ratio	0.83	1.01	1.01
Maximum drawdown	-13.04%	-11.24%	-15.57%
1-month 5% VaR	3.51%	3.30%	4.92%

Sharpe Ratio on returns in excess of the 1-month T-bill; t-statistic of the mean excess return = Sharpe × √ years. VaR = 1.645 × realised monthly volatility, as for the ex-ante VaR. Split tests the rule, with weights different from ours.

FIGURE 2 · GROWTH OF USD 1, MONTHLY REBALANCING



Full sample, Jul 2013 – Sep 2026, fixed weights estimated on Oct 2021 – Sep 2026 (light band). Equal weight and risk parity 1/σ are the comparisons of Checkpoint 2. [13, 14]

What the backtest tells us

TABLE 9 · CALENDAR-YEAR RETURNS

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026*
Our SAA	**0.8**	**-5.7**	**7.3**	**11.3**	**-4.8**	**14.3**	**11.9**	**6.7**	**-9.4**	**9.0**	**8.2**	**19.3**	**7.9**
Equal weight	0.3	-6.1	7.6	13.1	-5.7	15.3	12.7	7.3	-9.6	9.8	8.9	21.6	9.1
Risk parity 1/σ	2.5	-3.8	6.3	9.1	-3.4	12.7	10.9	4.7	-10.6	8.4	6.3	15.0	5.1

*Total return in %, USD. *January – September 2026. Fixed weights of section 06 for our SAA.*

7.2 · READING THE RESULTS

Return and risk. Over the full sample, USD 1 grew to USD 2.11: 5.9% a year with 7.4% volatility and a Sharpe Ratio of 0.55. The portfolio lost money in three of the twelve full calendar years (2015, 2018, 2022), never more than 9.4%.

2022, the stress test that matters. From October 2021 to September 2022 the portfolio lost 13.0%, its maximum drawdown. In 2022 US equity and US bonds fell together, so the usual hedge failed. Commodities gained and gold held roughly flat, which is the regime for which we hold them. The loss was recovered by March 2024. In March 2020 equities and commodities fell sharply, while bonds and gold held their value.

Volatility depends on the regime. With the same weights, realised volatility was 6.9% in 2013–2021, when stocks and bonds were negatively correlated, against 8.1% in the estimation window. Our weights are calibrated on the recent, less favourable regime, so they should not exceed the band if that regime persists.

The evidence is statistically weak. The t-statistics of the mean excess return are between 1.9 and 2.0, at the usual threshold of about 2: only the full sample, which is partly in-sample, just reaches it. This is the Class 2 lesson that decades of data are needed, and it is why Checkpoint 3 discounts the backtested Sharpe Ratio.

TABLE 10 · SHARPE RATIO AND EX-POST VOLATILITY OF THE ELEVEN ALLOCATIONS

	FULL SAMPLE		HOLD-OUT	
	SHARPE	VOL.	SHARPE	VOL.
CANDIDATES — BASKET METHOD				
Basket 33/33/33	0.50	6.46%	0.65	5.86%
Basket 50/25/25 · SAA	**0.55**	**7.39%**	**0.66**	**6.94%**
Basket 60/20/20	0.57	8.12%	0.66	7.79%
Basket 70/15/15	0.58	8.62%	0.66	8.41%
BENCHMARKS OF CHECKPOINT 2				
Equal weight	0.57	8.17%	0.64	7.81%
Risk parity 1/σ	0.49	6.15%	0.78	5.44%

	FULL SAMPLE		HOLD-OUT	
	SHARPE	VOL.	SHARPE	VOL.
OTHER METHODS — FOR COMPARISON				
Minimum variance	0.55	6.45%	0.64	6.11%
Maximum Sharpe	0.78	8.19%	0.64	7.85%
ERC, 8 ETFs	0.49	6.25%	0.68	5.69%
Mean–variance at 8%	0.77	7.60%	0.63	7.28%
Max. diversification	0.56	6.57%	0.60	6.31%

Ex-post volatility: realised, annualised. Full sample Jul 2013 – Sep 2026; hold-out Jul 2013 – Sep 2021.

Maximum Sharpe leads the full sample (0.78) only because its weights were fitted on the last 60 months; in the hold-out it falls to 0.64. In the hold-out, risk parity 1/σ has the highest Sharpe Ratio (0.78) and the lowest volatility (5.44%). Our basket is never the best nor the worst (0.55 in the full sample, 0.66 in the hold-out), and the four candidates have practically the same hold-out Sharpe Ratio (0.65–0.66). The hold-out has look-ahead bias, but it is the test of the mix we hold, not of the method and not of re-optimising.

Our allocation vs. the robust approaches

CP2.1 · WEIGHTS, RISK CONTRIBUTIONS AND VOLATILITY

EQUAL WEIGHT

$$w_i = 1 / N$$

RISK PARITY

$$w_i \propto 1 / \sigma_i$$

Both rules are applied to our eight ETFs, fully invested. The volatilities for risk parity, the risk contributions and the volatility of each portfolio use the same 60 months as our covariance matrix (October 2021 – September 2026).

	OUR SAA		EQUAL WEIGHT		RISK PARITY	
ETF	WEIGHT	RISK CONTR.	WEIGHT	RISK CONTR.	WEIGHT	RISK CONTR.
VTI	10.90%	15.21%	12.50%	16.43%	8.11%	12.69%
VEA	10.24%	17.42%	12.50%	19.97%	7.89%	14.90%
IEMG	10.23%	17.36%	12.50%	20.05%	7.59%	13.98%
BND	13.33%	8.34%	12.50%	6.98%	18.29%	14.35%
BNDX	16.48%	7.83%	12.50%	5.26%	22.20%	13.47%
TIP	15.26%	8.84%	12.50%	6.45%	20.62%	14.69%
GLD	11.15%	15.49%	12.50%	15.96%	7.24%	10.73%
BCI	12.40%	9.51%	12.50%	8.91%	8.05%	5.18%
BIL	0.01%	0.00%	0.00%	0.00%	0.00%	0.00%
Equity	31.4%	50.0%	37.5%	56.4%	23.6%	41.6%
Bonds	45.1%	25.0%	37.5%	18.7%	61.1%	42.5%
Real assets	23.5%	25.0%	25.0%	24.9%	15.3%	15.9%
Volatility	**8.13%**		**8.78%**		**7.23%**	

CP2.2 · WHERE THE RULES DISAGREE WITH OUR WEIGHTS

The three allocations share the same shape: every ETF holds 7–22% of the capital, all within the 7–9% band. They disagree in three places.

Equity: equal weight gives it 37.5% of the capital and 56% of the risk, because it ignores that equities are 2.6 times as volatile as bonds; risk parity gives it 24% and 42%. Our 31% of capital and exactly 50% of risk lies in between, and it is the one choice in our weights that rests on a view about returns. That view is a prior that the equity premium is the most reliable premium, not an estimated mean.

Bonds: risk parity puts 61% of the capital in the three bond ETFs because each looks safe on its own. But they are correlated 0.85–0.90 with each other and, since 2021, positively with equities, so $1/\sigma$ counts the same risk three times. Our ERC uses the correlations and holds 45%.

Real assets: risk parity cuts gold and commodities to 15% of the capital and 16% of the risk because they are volatile on their own. It ignores that they are the least correlated assets in the universe and the only ones that held up in 2022. We keep them at 25% of the risk, on the basis of correlations and our regime analysis, not their past returns.

None of our weights depends on an estimated expected return. We did not choose equal weight or risk parity $1/\sigma$ because they allocate risk ETF by ETF, not by asset class (section 5.2). The hold-out did not change our view: $1/\sigma$ had a higher Sharpe Ratio than ours (0.78 against 0.66) and equal weight a similar one (0.64), but eight years of data cannot separate Sharpe Ratios this close: each has a standard error of about 0.35 (CP3.4). Our weights rest on the design of section 05, not on backtested performance.

Backtest protocol

CP3.1 · DESIGN

Universe	8 ETFs + BIL. BCOMTR index for BCI over the whole sample (BCI listed 2017).
Weight rule	Basket method 50/25/25, ERC inside baskets, covariance of the last 60 months.
Rebalancing	Monthly, to the SAA weights; rest in BIL; no transaction costs.
Sample	Jul 2013 – Sep 2026, 159 monthly total returns (USD, Bloomberg); T-bill: FRED.

CP3.2 · VALIDATION

Tuning	Oct 2021 – Sep 2026 (60 months): the window that sets the weights.
Hold-out	Jul 2013 – Sep 2021 (99 months), never used for the weights. The split test uses only past data.
Variations	Up to **33** in sample: 11 allocations on three versions of the universe, all before any backtest. The 11 of the final universe were then run once on the hold-out.

The universe changed twice, both times before any backtest. VT, used while we wrote the code, was replaced by VTI + VEA + IEMG to control the regional mix and the US concentration (section 3.1). STIP was then replaced by TIP: its 0–5 year maturities gave the bond basket a short duration, against our intermediate target, and kept the portfolio's volatility low. In all, we had eleven portfolios in front of us: the four basket budgets we were choosing among, and the others we had committed not to choose (Checkpoint 2 benchmarks and methods we rejected on principle, section 05). In the hold-out, risk parity $1/\sigma$ beat our allocation (0.78 against 0.66) and we did not adopt it. We kept the hold-out out of the choice as far as we could: we chose the budget on the 8% target and looked at the hold-out only afterwards. It is still not fully clean, as a poor result would probably have made us revisit the choice (Class 4).

CP3.3 · RISK

	TUNING SAMPLE	HOLD-OUT SAMPLE	FULL SAMPLE
Mean excess return, annualised (t-stat.)	3.15% (0.87)	4.61% (1.91)	4.06% (2.00)
Volatility, annualised	8.13%	6.94%	7.39%
Maximum drawdown	-13.04%	-11.24%	-13.04%
One-month 5% VaR	3.86%	3.30%	3.51%
One-month 5% VaR, USD	3,860,471	3,295,225	3,508,981
Sharpe Ratio	0.39	0.66	0.55

Fixed SAA weights. VaR = 1.645 × realised monthly volatility (normal). Cumulative returns: Figure 2.

CP3.4 · THE DISCOUNT

BACKTESTED SHARPE (HOLD-OUT)

0.66

REALISTIC (DISCOUNTED)

0.33

We start from the hold-out, the only test of our weights on months they were not estimated on, and halve it (Class 4 rule of thumb). A raw 0.66 would overstate what we expect, for five reasons. (1) With up to 33 variations in sample, some look good by chance alone. (2) We looked at the hold-out before committing to the budget. (3) The ETFs were chosen in 2026, knowing that gold had rallied. (4) In 2013–2021 rates fell and stocks and bonds hedged each other, which no longer holds. (5) There are no trading costs. The t-statistic of 1.9 also means a standard error of about 0.35 on the Sharpe Ratio. We do not discount more because our weights have a single tuned parameter, the equity budget, and use no expected returns. This 0.33 is consistent with evidence that does not depend on our choices: in Class 3's 40-year horse race, risk parity earned 0.53 out of sample and a 60/40 portfolio 0.36.

The code

All the numbers in this report come from one Python notebook, attached to the report and published on our website. Running it from top to bottom reproduces every table.

SAA_R1.ipynb	§1–2 prices, T-bill and monthly returns; §3 the ETFs; §4 inputs on the last 60 months; §5 volatility, risk contributions and VaR; §6–7 the eleven allocations, ours included; §8 three backtests (full sample, hold-out, split); checkpoint cells for Checkpoints 1–3; §9 the shares for Bloomberg; §10 the other figures of the report.
AMC_adjusted_prices_final.csv	Bloomberg export: daily adjusted prices (dividends and splits included) of the eight ETFs, BIL and the BCOMTR index, January 2012 – 15 October 2026.
AMC_closing_raw.csv	Bloomberg export: raw closes of Friday 16 October 2026 of the eight ETFs and BIL, BCI included, used by §9.
FRED DGS1MO	Daily 1-month T-bill rate, downloaded by the notebook at every run (copy saved in DGS1MO.csv) and averaged by month.
portfolio_for_PRTU.csv	Output of §9: the 8 positions of the table below, entered in Bloomberg (PRTU).
report_figures.json	Output of §10: every number of this report, as printed here.

Portfolio description

We bought whole shares at the close of Friday 16 October 2026. For each ETF we bought as many shares as its target weight allows, rounding down. Rounding left some cash, including the optimiser's 0.01% in BIL; since cash earns nothing, we spent it on equity, the last USD 144.20 on two more VEA shares, the cheapest equity ETF. USD 5.22 remains in cash. The portfolio was entered in Bloomberg PRTU on 17–18 October and stays unchanged until Round 2.

SECURITY	BLOOMBERG TICKER	CCY	SHARES	PRICE	WEIGHT
Vanguard Total Stock Market ETF	VTI US Equity	USD	29,149	374.07	10.90%
Vanguard FTSE Developed Markets ETF	VEA US Equity	USD	147,369	69.49	10.24%
iShares Core MSCI Emerging Markets ETF	IEMG US Equity	USD	133,320	76.79	10.24%
Vanguard Total Bond Market ETF	BND US Equity	USD	188,225	70.82	13.33%
Vanguard Total International Bond ETF	BNDX US Equity	USD	349,731	47.13	16.48%
iShares TIPS Bond ETF	TIP US Equity	USD	144,602	105.54	15.26%
SPDR Gold Shares	GLD US Equity	USD	29,177	382.01	11.15%
abrdn Bloomberg All Commodity Strategy K-1 Free ETF	BCI US Equity	USD	469,258	26.42	12.40%
Cash, rounding leftover (USD 5.22)		USD			0.00%
Total (USD 100,000,000)		**USD**			**100.00%**

Price date: close of Friday 16 October 2026 for every position. Actual weights after rounding: at most 0.004 percentage points from the targets of section 06 (equity slightly above). Portfolio held: ex-ante volatility 8.13%, 1-month 5% VaR 3.86% = USD 3,861,174.

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[14] Federal Reserve Bank of St. Louis, FRED: DGS1MO, Market Yield on U.S. Treasury Securities at 1-Month Constant Maturity, downloaded by the notebook at each run (copy saved in DGS1MO.csv).
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